Analytics Engineering: Session 15

Certification Prep & Mock Exam

Agenda

- Certification Exam Overview
- Domain-by-Domain Quick Review
- Mock Exam (Timed)
- Group Answer Review
- Targeted Weak-Area Review
- Next Steps

Certification Exam Overview

- **Format**: 65 Multiple Choice Questions.
- **Time**: 2 Hours.
- **Passing**: 65% (43 correct answers).
- **Focus**: Practical application, not just theory.
- **Version**: dbt 1.7+

Strategy:

- Read each question carefully — look for keywords like "always", "never", "best".
- Flag and skip difficult questions — come back to them.
- Manage your time — ~1.8 minutes per question.

Domain 1: Developing dbt Models

Key topics:

- `ref()` and `source()` functions
- Core materializations (view, table, ephemeral, incremental)
- Modularity, DRY, CTEs
- DAG structure and execution order
- `dbt_project.yml` configuration
- dbt commands: `run`, `test`, `build`, `seed`, `docs`
- Python models
- `grants` configuration

Sample question: *"What happens if Model A uses `ref('Model B')` and Model B uses `ref('Model A')`?"*

Domain 2: Model Governance

Key topics:

- Contracts: enforcing schema at build time
- Versions: managing breaking changes
- Deprecation: signaling end-of-life
- Access levels: public, protected, private

Sample question: *"If a model has* `contract: {enforced: true}` *and a column returns* `varchar` *instead of* `int`*, what happens?"*

→ The build fails with a contract violation error.

Domain 3: Debugging Errors

Key topics:

- Reading logged error messages (bottom up)
- Troubleshooting with compiled SQL (`target/compiled/`)
- YAML compilation errors (indentation, syntax)
- Distinguishing dbt errors from database errors
- Developing and testing fixes before merging

Sample question: *"A model fails with 'relation does not exist'. Where should you look first?"*

→ Check the `ref()` or `source()` name for typos, then check compiled SQL.

Domain 4: Managing Data Pipelines

Key topics:

- Managing DAG failure points
- `dbt clone` for environment management
- Troubleshooting errors from integrated tools

Sample question: *"After a partial pipeline failure, what is the most efficient way to resume?"*

→ Use `dbt retry` to re-run only the failed nodes and their downstream.

Domain 5: Implementing dbt Tests

Key topics:

- Generic tests: `unique`, `not_null`, `accepted_values`, `relationships`
- Singular tests: custom SQL in `tests/`
- Custom generic tests: reusable macros in `macros/`
- Testing sources and models
- `dbt build` runs tests in DAG order

Sample question: *"What is the difference between a singular test and a custom generic test?"*

→ Singular tests are one-off SQL files. Custom generic tests are reusable Jinja macros that can be applied to any model via YAML.

Domain 6: Documentation

Key topics:

- Adding descriptions in YAML files
- Doc blocks with `{% docs %}` in markdown
- `dbt docs generate` and `dbt docs serve`
- Source, model, and column documentation

Sample question: *"How do you reference a doc block in a model's YAML description?"*

→ `description: '{{ doc("my_doc_block_name") }}'`

Domain 7: External Dependencies

Key topics:

- Exposures: defining downstream consumers (dashboards, reports)
- Source freshness: `loaded_at_field`, `warn_after`, `error_after`

Sample question: *"What is the purpose of defining an exposure in dbt?"*

→ Exposures document downstream consumers (dashboards, ML models) and make them visible in the lineage graph for impact analysis.

Domain 8: Leveraging dbt State

Key topics:

- State artifacts: `manifest.json`
- `state:modified` and `state:new` selectors
- `dbt retry` for re-running failed nodes
- Result selectors: `result:error`, `result:fail`, `result:pass`

Sample question: *"What is the difference between `state:modified` and `result:error`?"*

→ `state:modified` compares current code to a previous manifest. `result:error` looks at the results of the last run to find nodes that errored.

Mock Exam

Instructions:

- 30 questions, 50 minutes (simulating exam pace)
- Work individually — no notes or docs
- Mark questions you're unsure about
- We'll review answers together afterward

Starting now...

Mock Exam Review

Let's go through the answers together.

For each question:

1. What is the correct answer?
2. Why are the other options wrong?
3. What concept does this test?

Focus on understanding the "why", not just memorizing answers.

Targeted Review

Based on the mock exam results, let's focus on the areas where the group scored lowest.

Common weak areas:

- Incremental strategies and `is_incremental()` logic
- Snapshot configuration details (dbt-added columns)
- State vs result selectors
- Contract constraint types
- `dbt build` vs `dbt run` vs `dbt test`

Exam Day Tips

1. **Read carefully**: Watch for "BEST", "ALWAYS", "NEVER" in questions.
2. **Eliminate**: Remove obviously wrong answers first.
3. **Time management**: Don't spend more than 2 minutes on any question.
4. **Flag and return**: Mark uncertain questions and come back.
5. **Compiled SQL**: Many questions test whether you understand what dbt sends to the database.
6. **Think in DAGs**: Model execution order, downstream impact, selectors.

Next Steps

1. **Take the exam**: getdbt.com/certifications
2. **Study resources**:
   - dbt Fundamentals course
   - Official Study Guide (PDF)
   - dbt Documentation
3. **Practice**: Build your own dbt project with a different dataset.
4. **Community**: Join the dbt Slack for support.

Good Luck!

You are now ready to be a **dbt Certified Analytics Engineer**.

What have we learned in this session

- Reviewed all 8 certification domains with sample questions
- Completed a timed mock exam
- Identified and reviewed weak areas
- Exam strategy and time management tips

End of Course.